

Project Chronos – Technical Architecture Overview

1. Serverless Architecture Design

Project Chronos implements a fully serverless architecture leveraging AWS Lambda for compute execution, Amazon API Gateway for secure API exposure, and Amazon DynamoDB for highly scalable NoSQL data persistence. The system is deployed inside a Private VPC Subnet, ensuring no direct public network exposure. Service-to-service communication is routed through AWS PrivateLink, eliminating internet traversal and reducing attack surfaces. All data at rest and in transit is encrypted using AWS KMS-managed encryption keys to enforce enterprise-grade security controls.

2. Security & Compliance Controls

The platform enforces least-privilege IAM roles, encrypted DynamoDB tables using AWS KMS, private API Gateway endpoints, and VPC endpoint policies to restrict service communication. Audit logs are captured via AWS CloudTrail and CloudWatch.

3. Data Flow Diagram

